

# DASHI: A Fail-Closed Constructive Physics-Unification Architecture

Carrier Geometry, Filtration, Projection-Defect Structure, and Mechanized Closure Frontiers

DASHI project

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**Core formalism.** Paper 1 uses the following grammar to stage DASHI’s current physics-unification carrier without treating the grammar itself as a universal physical law:

$$x \in \mathcal{X} \simeq (a, \nu, \tau, d, \ell, r), \quad \tau \in \{-1, 0, +1\}.$$

Here  $a$  is a prime/supersingular-prime address,  $\nu$  is **FactorVec**-style valuation data,  $\tau$  is the ternary coordinate,  $d$  is refinement depth,  $\ell$  is the active lane, and  $r$  records receipt/projection status. A surface  $s$  exposes only its admitted projection:

$$x = P_s(x) + D_s(x), \quad \rho(x, y) = \alpha^{k(x, y)}, \quad 0 < \alpha < 1.$$

$P_s(x)$  is the promoted component;  $D_s(x)$  is the residual defect; and  $k(x, y)$  is the first incompatible address, valuation, or prefix depth. Filtered operator surfaces use  $F_0 \subseteq F_1 \subseteq \dots$ ,  $\text{gr}_n F = F_n / F_{n-1}$ , and descended brackets only where quotient, equivalence, norm, and bracket laws are inhabited. Lane actions commute only on the official tracked independent **GL.FactorVec** route. UFT/motif compression supplies reusable projected structure while residual defects remain visible.

## Abstract

DASHI is a constructive physics-unification program built around a typed carrier geometry. Its central object is not a single fitted equation, but a coherent mathematical substrate in which prime-coordinate carriers, ultrametric refinement, projection-defect decomposition, filtered operator surfaces, lane independence, local root-geometry boundaries, compression semantics, and empirical-contact receipts can be staged inside one typed architecture.

The manuscript uses a projection-defect schema to organize promoted content and residual obstruction across those surfaces. Filtrations organize limiting behavior, ultrametric addresses organize refinement depth, lane actions organize independent coordinate sectors, and admissibility gates determine which constructions may be read as physics-facing claims. Typed closure and obstruction semantics are therefore not external project governance; they are the enforcement layer of the unification substrate itself.

Paper 1 presents the current constructive frontier of this program. The positive inventory includes direction-indexed G2 SFGC schema adoption, an official tracked G6 cross-lane commuting route, constructive finite-support G3 selected-operator subtraction support, an E8/LILA integer and local semantic boundary, a typed cross-domain projection-defect / MDL spine boundary, and an inhabited bounded W3 below-Z Drell-Yan  $\mathfrak{t}43$  comparison-law receipt with  $\text{chi2/dof} = 2.1565191176$  under threshold 4.0. The contribution is a mathematically coherent ultrametric carrier architecture for physics unification whose closure semantics expose the exact frontier rather than hide it.

# 1 Introduction

The starting point of DASHI is that a physics-unification program needs a carrier rich enough to hold geometry, dynamics, compression, empirical contact, and unresolved structure without forcing them into the same proof status. The basic pattern is a state carried by prime/trit and **FactorVec**-style coordinates, refined by ultrametric depth, acted on by lane-local operators, and read through projections whose residuals remain visible. This is a physics thesis before it is a governance thesis: the closure discipline is needed because the proposed carrier is ambitious, not because the paper is about process management.

The organizing picture is projection from structured possibility into a surface-visible form. A candidate state can carry more information than a particular physics-facing surface is allowed to expose. The promoted projection is the piece that survives the current admissibility gate; the defect records what is left unresolved, incompatible, or not yet transported across the relevant boundary. Filtration gives limiting and contraction language, ultrametric refinement gives address and precision language, and lane independence gives compositionality between coordinate sectors.

This architecture lets several mathematical neighborhoods meet without being collapsed into metaphor. Direction-indexed discrete sites supply local carrier geometry. Finite-support operator surfaces supply controlled algebra for filtered contraction targets. Tracked cross-lane commuting supplies a coordinate-independence law for selected lane actions. Local E8/LILA surfaces supply finite root-geometry and upstream-promotion boundaries. UFT-style compression supplies refinement, motif, and residual language. The Drell-Yan lane supplies a bounded example of empirical contact bound to a frozen typed receipt.

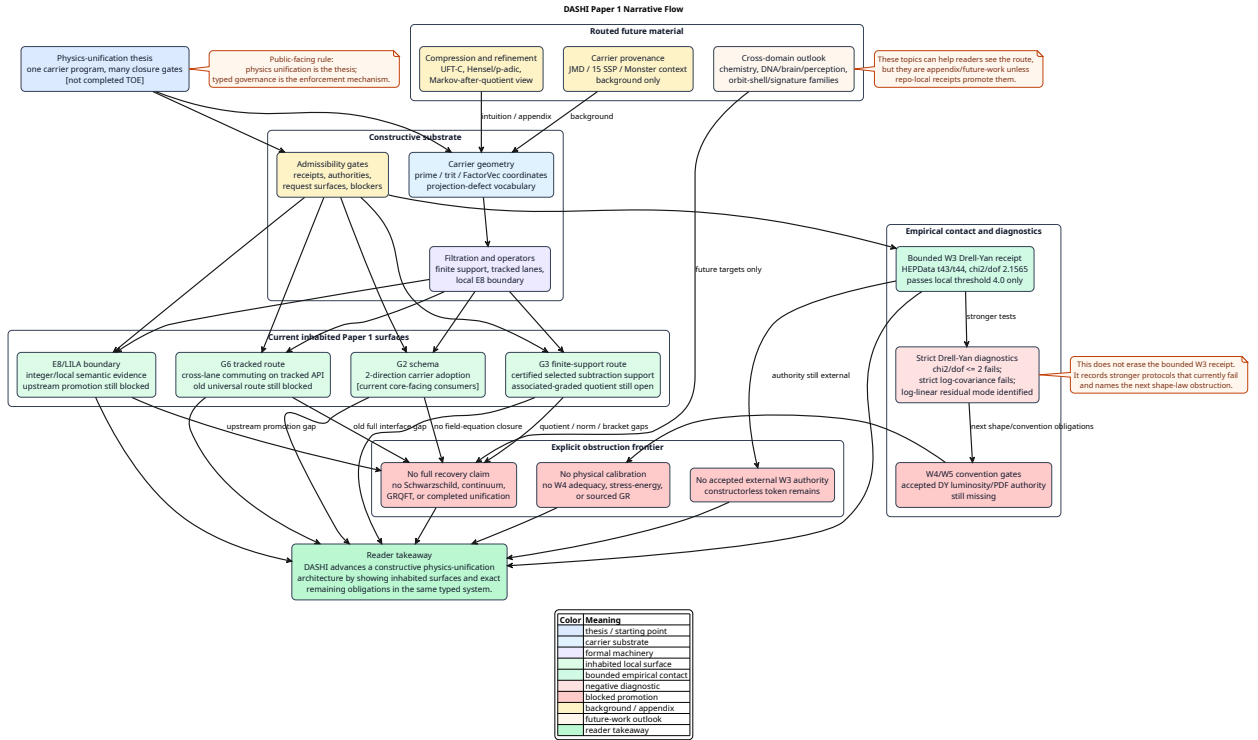


Figure 1: Reader-facing narrative flow for Paper 1. The diagram foregrounds the constructive carrier substrate, local mathematical surfaces, empirical contact, compression geometry, and the later certification frontier. It is a narrative map, not a proof of completed unification.

## 2 Background and Related Work

DASHI draws on standard mathematical neighborhoods including ultrametric and p-adic structure, lattice and gauge language, contraction and associated graded constructions, root-system classification, mechanized mathematics, and compression-style trie geometry. The claims made here are restricted to the repo-local receipt and obstruction surfaces cited in the claim ledger.

The formal-methods positioning is central. DASHI uses Agda as a typed development surface: selected propositions, receipts, audits, and blockers are represented as typed artifacts, and the absence of a required inhabitant is manuscript-relevant state. This places the paper near proof assistants, dependent type theory, mechanized mathematics, and proof-carrying artifacts, while remaining narrower than an end-to-end certified physics pipeline.

Ultrametric and p-adic language organizes refinement by depth: nearby states share high-level valuation or prefix structure, while separation can occur at the first incompatible digit or coordinate. Hensel-style lifting is used only as an external analogy for compatible refinement across precision levels. In Paper 1 this is expository support for carrier and compression geometry, not an independent proof of convergence.

## 3 From Ternary Traversal to Carrier Geometry

The origin path into DASHI is best read as an iterative formalization of traversal: a state is first treated as a trit-valued position, then as a refinable address, then as a carrier point whose projections and residuals are tracked by typed operators. The point of the 3-6-9 or Base369 vocabulary is not numerological. Its safe role is to supply cyclic finite carriers: triadic, hexadic, and nonary state spaces with rotations, wrapped addition, and quotient/refinement seams. At the lowest level, a trit distinguishes three statuses: positive, negative, and unresolved or neutral.

Older dialectical language in the archive should be read in the same disciplined way. A dialectical state may be carried through a transition while retaining unresolved residue. A counter-state, conflict state, or overflow state is not a mystical third force; it is a typed indication that the present projection has not discharged all of the structure in the carrier. In the current manuscript this becomes the projection-defect discipline: a selected surface exposes  $P_s(x)$ , while the residual  $D_s(x)$  records what is blocked, incompatible, untransported, or not yet promoted.

The geometric intuition then moves from a one-dimensional cycle to traversal through a cube-like state space. Rubik-like cube, hypercube, and voxel language is useful only as intuition for finite-coordinate movement: each coordinate records a local trit, phase, or valuation, and a traversal is a sequence of admissible coordinate updates. Voxel language fits the same role at nonary scale: a local  $3^2$  or  $3^k$  block can be read as a small cell of structured state, and overflow means that the current cell cannot preserve the relevant invariant without adding a refinement coordinate.

Video and state-continuity testing belong at this transition point as engineering diagnostics. A video trace is a sampled path through state space: continuity testing asks whether adjacent frames preserve the expected local carrier relation, whether an apparent jump is an admissible carry/refinement, and whether the projection defect increases at the boundary. Such tests can motivate carrier choices and expose discontinuities, but they are not theorem authority unless the trace, observation map, metric, and admissibility law are bound by a repo-local receipt. The sibling **dashifine** and **dashitest** repositories contain concrete video-side evidence at this engineering level: GIF/frame visualizations for closure and ultrametric diagnostics, codec and Vulkan/VA-API video benchmark paths, live-sheet recording scripts, and short MP4 smoke artifacts. These may be cited as sibling diagnostic evidence, not as DASHI theorem authority, until ported into a local receipt

with command and artifact digests.

UFT/trie refinement gives this traversal a compression geometry. A trie node is a coarse address; descending the trie adds precision. Nearby branches share prefix structure, and separation occurs at the first incompatible depth. This matches the ternary ultrametric reading already used in the carrier spine: refinement depth is not just a storage index but a metric and semantic relation. A motif is then a reusable local shape in the trie, not an equality claim.

**FactorVec** is the corresponding prime-coordinate refinement of the same idea. Instead of only trit positions, a state carries valuation data across tracked prime lanes. Multi-lane transport, chamber legality, signed scans, and defect summaries turn traversal into a controlled carrier geometry: a move may be legal or illegal, stable or repatterning, contractive or expansive, and these labels are computed at the representation layer rather than inferred from prose.

This origin story gives a disciplined reading of the transition from trits to carrier geometry. DASHI starts with ternary traversal because trits make unresolvedness explicit. It passes through **Base369**, cube/voxel traversal, video-like continuity traces, UFT tries, and **FactorVec** lanes because each adds a more precise way to talk about transition, address, refinement, and defect. None of this proves full physics unification, automatic convergence, Monster representation semantics, empirical adequacy, or domain recovery. It supplies a traceable formal path by which those stronger claims would have to be typed, projected, tested, and closed.

## 4 Why the Carrier Geometry Is Forced

The carrier is not chosen by taste. It is the smallest structure that keeps four pressures visible at once: unresolvedness, refinement, locality, and projection. If a state can be affirmed, denied, or still undischarged by the current surface, binary truth is already too small. A ternary coordinate is the first stable carrier for that situation: +1, -1, and 0 are not metaphysical labels, but bookkeeping for oriented evidence and unresolved residue.

Once unresolved state is allowed to refine, a tree appears almost immediately. A coarse node records what is visible at the current depth; descending the trie adds distinctions. Two states that share a long prefix are close for any observer or surface that cannot see past that depth, and the first incompatible digit is the natural separation event. This is why the ultrametric is not decoration. It is the metric forced by recursive refinement: distance is controlled by the earliest depth at which two histories, addresses, or motifs can no longer be jointly refined.

Prime-coordinate and **FactorVec** structure enter when refinement must also remember independent axes of change. A factorized carrier records how much of each tracked prime lane is present. If one update changes only the **p** valuation and another changes only the **q** valuation, then for  $\mathbf{p} \neq \mathbf{q}$  the two updates should commute on the tracked projection. G6 is the formal version of that intuition: independent valuation coordinates naturally commute, but only the official tracked theorem is allowed to say so inside the paper.

Projection-defect decomposition is then inevitable rather than decorative. Every observation, theorem surface, empirical receipt, or compression motif exposes only part of the carrier. A less disciplined theory would forget the remainder and later reintroduce it as an assumption. DASHI keeps the remainder as  $D_s(x)$ : the residual defect relative to a surface. This is why obstruction belongs to the mathematics. It is the unpromoted part of the carrier after projection, not a separate editorial apology.

Filtration is the same idea applied to operators. A local move is useful only if it respects the depth and quotient structure on which the carrier is being read. When an operation preserves the relevant filtration and descends to the quotient, it can become a theorem-facing action on the

associated graded surface. When it does not, the failure is visible as a missing equivalence, missing quotient carrier, or missing descended law. G3 is therefore not just a blocker list; it is the exact place where finite-support operator intuition is being asked to become quotient-correct algebra.

The root-geometry and empirical lanes fit the same pattern. E8/LILA supplies finite local shape and symmetry boundaries where enumeration and promotion can be audited. The Drell-Yan lane supplies a bounded observable projection where a frozen comparison can be checked while stricter residual shape laws remain visible. These are different projections of the same recursively refining carrier grammar. Paper 1 is compelling only if the reader can see that unity before seeing the blocker table.

## 5 Core Constructive Architecture

The core DASHI architecture is a carrier geometry before it is a list of gates. The carrier is intended to hold prime-coordinate location, trit-valued or unresolved state, **FactorVec**-style valuation data, lane-local actions, depth/refinement information, and typed projections into the surfaces a reader can inspect. This is the mathematical reason the architecture can talk about discrete gauge sites, filtered operators, E8/LILA root boundaries, compression motifs, and empirical comparison receipts without treating them as unrelated subsystems.

### 5.1 Carrier Coordinates and Depth

The paper-facing carrier intuition is coordinate-rich but not continuously overcommitted. Prime and supersingular-prime coordinates provide addressable axes. Trit and Base369-style language supplies a compact way to discuss positive, negative, and unresolved or neutral positions. **FactorVec** surfaces make valuation information explicit enough for tracked lane actions. UFT-style addresses add a refinement geometry: nearby addresses share high-level prefix or valuation structure, and separation occurs at the first incompatible depth.

This gives the carrier a natural physics reading. A state is not merely a point in a set; it is a structured address with depth, projection grade, valuation content, and possible unresolved branches. Refinement can therefore be discussed without pretending that every refinement is already an observable physical fact. Hensel-style language is useful as intuition for compatible partial information lifting through increasing precision.

### 5.2 Projection-Defect Decomposition

The central local equation is the projection-defect split:

$$x = P(x) + D(x).$$

Here  $P(x)$  denotes the promoted projection visible to a chosen surface, and  $D(x)$  denotes the residual defect left by that projection. The point is not that every domain has the same observables. The point is that DASHI repeatedly uses the same structural move: expose the admissible part of a richer carrier, preserve the residual, and make later promotion depend on a typed bridge rather than a rhetorical analogy.

### 5.3 Filtration, Lane Action, and Compression

Filtration supplies the limiting direction of the architecture. It lets DASHI ask whether a restricted operator surface descends to an associated graded object, whether contraction language is quotient-correct, and whether norms or brackets survive projection. Lane actions supply coordinate-local

dynamics: selected transformations can act on tracked prime sectors, and the official G6 theorem checks that selected cross-lane actions commute on the tracked route.

Compression is not decorative. The UFT and motif vocabulary supply a way to talk about reusable structure, address refinement, branch compatibility, and semantic distance. A motif is not equality; it is evidence of reusable projection structure at a chosen grade. This is why compression belongs in the physics paper: it is the language by which a rich carrier becomes searchable, refinable, and composable without collapsing residual structure.

## 5.4 Closure Semantics as Enforcement

The closure layer enters after the carrier has something worth protecting. Fail-closed promotion says that a local theorem, audit, or receipt promotes only the claim authorized by its owning surface. A request surface can point to an empirical, geometric, or upstream mathematical authority, but it does not supply that authority. This is the enforcement architecture of the unification program: it keeps the carrier ambitious while making the claim surface exact.

## 6 Mathematical Constructions

| Label   | Mathematical role                                   | Current positive surface                                                         | Boundary                                                          |
|---------|-----------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------|
| G2      | Direction-indexed discrete carrier schema           | SFGC schema adoption for current core-facing consumers                           | No Maxwell closure or field-equation recovery follows             |
| G3      | Restricted finite-support operator filtration       | Certified subtraction support for selected constructive finite-support operators | Associated-graded quotient infrastructure remains missing         |
| G6      | Tracked cross-lane coordinate independence          | Official tracked cross-lane commuting theorem and above-threshold independence   | Old universal <code>LaneOperator</code> promotion remains blocked |
| E8/LILA | Local finite root-enumeration and upstream boundary | Integer/two-sparse completeness plus local semantic audit boundary               | Upstream <code>E8RootEnumerationComplete</code> remains blocked   |

### 6.1 G2: Direction-Indexed Discrete Gauge Schema

The G2 lane supplies the current direction-indexed carrier schema for local discrete gauge geometry. The positive surface is `canonicalG2SFGCGaugeFieldSchemaExtensionReceipt`, with the no-remaining-missing surface `noRemainingG2SFGCSchemaExtensionMissing`. The admissible paper claim is narrow: direction-indexed SFGC schema adoption is complete for the current core-facing consumers. This does not prove Maxwell closure, Yang-Mills dynamics, field-equation recovery, empirical calibration, limited Standard Model plus GR closure, GRQFT closure, or completed unification.

### 6.2 G3: Selected Finite-Support Operators and the Quotient Gap

The G3 lane is the restricted operator-calculus lane. The current positive result is certified subtraction support for a selected constructive finite-support

subtype, through `selectedFiniteSupportOperatorSubtractionCertifiedSupport`, `selectedFiniteSupportOperatorSubtractionSupportWitnessFromCertified`, and `G3SelectedSubtractionCertifiedSupport`. A genuine associated graded layer still requires quotient classes of the form  $F_n / F_{n-1}$  and the kernel-equivalence law:

$$[x] = [0] \iff x \text{ lies in the denominator filtration piece.}$$

Until the accepted equivalence, quotient carrier, projection-respects-equivalence, and descended norm/bracket/isomorphism laws are inhabited, Paper 1 cannot claim global Schrodinger promotion, arbitrary operator finite support, norm-law closure, product support, bracket support, or a completed Galilei contraction theorem.

### 6.3 G6: Official Tracked Cross-Lane Commuting

The G6 lane records cross-lane independence inside the prime-coordinate carrier. An update to the `p` lane and an update to the `q` lane factor through distinct valuation exponents, so for `p != q` the order of those two tracked updates should not change the projected state. The theorem matters because DASHI does not leave that independence as prose; it pins it to the official tracked `GL.FactorVec` route. The positive surfaces are `canonicalG60officialTrackedCrossLaneCommutingTheorem` and `canonicalG60officialTrackedAboveThresholdCoordinateIndependence`. The old full `LaneOperator` interface is not promoted; its universal vanished-prime identity law blocks the non-trivial scaling route used by the tracked theorem. Paper 1 may claim the official tracked cross-lane commuting theorem and tracked above-threshold coordinate independence, but not old universal `LaneOperator` promotion, full G6 closure, or downstream physics closure from G6 alone.

### 6.4 E8/LILA: Local Completeness and Upstream Promotion Boundary

The E8/LILA lane is a finite root-enumeration and promotion boundary, not a completed E8-to-physics derivation. The positive and boundary surfaces are `integerIndexedRootsCompleteForTwoSparseShapeTheorem`, `canonicalE8RootEnumerationCompletePromotionBoundary`, and `canonicalE8UpstreamCompleteReceiptPromotion`. Paper 1 may describe local integer completeness and the exact upstream-promotion audit. It must not claim upstream `E8RootEnumerationComplete`, E8-derived Standard Model recovery, limited SM+GR, GRQFT closure, or completed unification from E8 structure.

## 7 Empirical Contact

The empirical-contact lane is intentionally narrow. Its role is to show that the constructive architecture can bind a named physics observable to a frozen, mechanically tracked receipt while still blocking stronger claims whose authority, convention, or strict-fit obligations are missing.

The current positive surface is the bounded W3 below-Z Drell-Yan comparison-law receipt over CMS-SMP-20-003 / HEPData `ins2079374/t43`, using the `ins2079374/t44` covariance convention. The receipt records `chi2/dof = 2.1565191176`, threshold 4.0, and mean prediction/data 0.9941233097. It has no above-Z or accepted-authority promotion.

Two stricter protocols are explicitly negative. The adjacent-ratio receipt records that the stricter `chi2/dof <= 2` falsification-lane target is not satisfied. The strict log-covariance diagnostic is also negative: `phiStarRatioPredictor` has strict log `chi2/dof = 283.45739523864586`, and `sigmaDashiv4Predictor` has strict log `chi2/dof = 3180.211733150705`. The main obstruction

is the log-linear residual subspace; the next theorem target is a frozen no-refit route discharging `StrictPassOrthogonalityObligation`, not a prose promotion.

The bounded W3 comparison-law receipt is not accepted W3 empirical authority. No accepted provider token, non-postulated policy hook, or governance payload currently inhabits `W3AcceptedEvidenceAuthorityToken`. W4 and W5 remain blocked by missing accepted Drell-Yan luminosity/convention authority.

## 8 Compression, Semantic Geometry, and Cross-Domain Spine

The UFT vocabulary supplies a compression-facing view of the carrier: addresses refine ultrametrically, motif membership is graded by projection evidence, and unresolved or incompatible branches remain represented rather than collapsed. The visual intuition is a floating trie of possible refinements. A coarse address names a region of structure; descending the trie adds precision, distinguishes branches, and exposes incompatibilities. Unlike an ordinary lookup tree, the UFT reading is semantic and typed: branch distance is ultrametric, branch membership is projection-grade dependent, and unresolved branches can remain live without being forced into a binary collapse.

Motifs then become reusable local shapes inside that trie. A motif does not assert identity between two states. It says that two carrier regions share enough projection structure to support a typed comparison at a chosen grade. Exact members, compatible members, blocked members, and residual members are different mathematical statuses. This lets the architecture reuse structure without treating resemblance as theoremhood.

UFT-C, lattice-logic, null/hinge codes, SWAR implementation machinery, Base369/trit notation, Hensel-style refinement, and Markov-after-quotient readings are routed as background or appendix material unless a repo-local surface promotes a stronger claim.

The cross-domain material is included as a theorem-target boundary, not as a new promoted result. The current Agda surface `canonicalCrossDomainVariationalSpineBoundary` records a common typed object (`X`, `delta`, `pi`, `defect`, `gate`, `observation`, `symmetry`) shared by physics, molecular, bonding, biological, and perceptual target rows. Its value for Paper 1 is structural: it shows how projection, defect, admissible descent, observation quotient, and symmetry-orbit language can be stated in one typed grammar. It does not imply chemistry closure, bonding-energy prediction, molecular spectra, protein folding, biological prediction, perceptual empirical fit, or universal TOE closure.

## 9 Obstruction Frontier

The preceding sections put the carrier first: coordinate structure, projection-defect decomposition, filtration, lane action, compression, local mathematical surfaces, empirical contact, and cross-domain spine. The obstruction frontier now records the exact residuals of that architecture. It is not the emotional center of the paper, but it is the certification layer that makes the physics-unification thesis admissible.

| Gate | Current surface                      |      | First missing<br>token                                   | primi-<br>token | Forbidden promo-<br>tion               |
|------|--------------------------------------|------|----------------------------------------------------------|-----------------|----------------------------------------|
| G2   | Direction-indexed<br>schema adoption | SFGC | Maxwell/curvature/field-<br>equation carrier<br>receipts | and             | Maxwell closure,<br>physical curvature |

|                    |                                             |                                                          |                                              |
|--------------------|---------------------------------------------|----------------------------------------------------------|----------------------------------------------|
| G3                 | Selected finite-support subtraction support | Quotient carrier, kernel-equivalence law, descended laws | Global operator or Schrodinger closure       |
| G6                 | Official tracked commuting route            | Migration or split of old universal law                  | Old full <del>LaneOperator</del> promotion   |
| E8/LILA            | Local integer/root boundary                 | Upstream completion constructor/API                      | Upstream E8 completion                       |
| W3                 | Bounded Drell-Yan receipt                   | <del>W3AcceptedEvidenceAuthorityToken</del>              | <del>Accepted empirical</del> authority      |
| W4/W5              | Request/diagnostic surfaces                 | Accepted DY convention and correction receipts           | Matter field, stress-energy, sourced physics |
| GR                 | Future adapter path                         | Non-flat connection, curvature, Ricci, sourced interface | Einstein tensor, Schwarzschild, continuum GR |
| Cross-domain spine | Shared projection-defect / MDL schema       | Calibration, universality, domain receipts               | Chemistry/biology/perception recovery        |
| Provenance         | JMD/SSP/archive/source routing              | Repo-local receipts or accepted authority                | Provenance-as-proof                          |

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The frontier has a constructive interpretation. Every row identifies a residual defect: the exact bridge, authority, quotient law, adapter, or calibration object that would be needed for a stronger projection. This is how DASHI avoids converting an elegant carrier into an overclaimed physics result.

## 10 Reproducibility

The canonical authoring surface is `Docs/PaperDraftWorkingFolder/Paper1_Manuscript.md`. The claim-control surface is `Docs/PaperDraftWorkingFolder/ClaimLedger.md`, and source/provenance routing is recorded in `Docs/PaperDraftWorkingFolder/SourceLedger.md`, `ArchiveTopicIntegrationPlan.md`, and `ArchiveRelativeLocationReference.md`.

The narrow verification surface for this draft is:

```
plantuml -checkonly Docs/*.puml
./scripts/render_docs_diagrams.sh
git diff --check -- Docs/PaperDraftWorkingFolder Docs README.md
```

## 11 What This Paper Does Not Claim

This paper does not claim that DASHI has completed physics unification. It does not claim an accepted theory of everything, accepted external empirical authority, physical calibration, sourced Einstein equations, non-flat GR closure, Schwarzschild recovery, continuum recovery, GRQFT closure, limited SM+GR closure, strict Drell-Yan success, above-Z Drell-Yan promotion, W4/W5 authority closure, upstream E8 completion, universal G6 closure, arbitrary wavefunction-operator closure, chemistry closure, biology prediction, perceptual validation, or cross-domain recovery equivalence.

## 12 Conclusion

DASHI should be read as a constructive physics-unification program centered on a coherent carrier geometry. Prime-coordinate structure, ultrametric refinement, projection-defect decomposition, filtered operator surfaces, tracked lane actions, local root-geometry boundaries, semantic compression, and bounded empirical receipts are staged as parts of one mathematical architecture. The paper's thesis is that this architecture makes unification a constructive closure problem rather than a sequence of informal analogies.

The current frontier is substantial but not final. G2 supplies a direction-indexed carrier schema for current core-facing consumers. G3 supplies selected finite-support subtraction support and exposes the associated-graded quotient gap. G6 supplies the official tracked cross-lane commuting route. E8/LILA supplies local integer/root-boundary structure while preserving upstream ownership. UFT and motif language supply the compression geometry that keeps refinement, semantic distance, and residual structure in the same typed vocabulary. The Drell-Yan lane supplies bounded empirical contact and the stricter residual shape law that still fails.

Typed closure is what keeps this architecture from becoming another unification narrative closed by assertion. It does not reduce the physics ambition; it makes the ambition legible. The positive surfaces stand where they are inhabited, and the stronger empirical, GR, upstream E8, universal-operator, and completed-unification readings remain exact residual obligations. Paper 1's contribution is therefore a mathematically coherent ultrametric carrier architecture for physics unification with a visible, mechanized closure frontier.

## References

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